Statistics for Data Science - 2

Week 7 practice Assignment

Statistics from samples and Limit theorems

1. A population has mean 60 and standard deviation 6. Random samples of size 100 from this population are collected independently. Find the expected value of the sample mean.
Solution:
We know that expected value of the sample mean $\overline{X}$ is given by

$$\begin{aligned} E[\overline{X}] &= \mu \\ &= 60 \end{aligned}$$

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2. Let X_1, X_2, X_3, X_4 and $X_5 \sim$ i.i.d. Normal$(2, 25)$. Calculate $P(2X_1 + X_2 + 3X_3 + X_4 + X_5 \geq 10)$.

   1. $F_Z(0.3)$
   2. $1 - F_Z(0.3)$
   3. $F_Z(-0.3)$
   4. $1 - F_Z(-0.3)$

   Solution:

   We know that linear combination of independent Normal distributions is again a normal distribution.

   Hence, $2X_1 + X_2 + 3X_3 + X_4 + X_5$ will follow a Normal distribution.
   Let $Y = 2X_1 + X_2 + 3X_3 + X_4 + X_5$
   $E[Y] = E[2X_1 + X_2 + 3X_3 + X_4 + X_5] = (2 + 1 + 3 + 1 + 1)E[X] = 16$
   $\text{Var}(Y) = \text{Var}(2X_1 + X_2 + 3X_3 + X_4 + X_5) = (4 + 1 + 9 + 1 + 1)\text{Var}(X) = 400$

   It implies that $Y \sim$ Normal$(16, 20^2)$.
   To find: $P(Y \geq 10)$

Now,

$$\begin{aligned}
P(Y \geq 10) &= P(Y - 16 \geq -6) \\
&= P(\frac{Y-16}{20} \geq \frac{-6}{20}) \\
&= P(\frac{Y-16}{20} \geq -0.3) \\
&= P(Z \geq -0.3) \\
&= 1 - P(Z < -0.3) \\
&= 1 - F_Z(-0.3)
\end{aligned}$$

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3. Random samples of size 100 are collected from a population of unknown parameters. If the variance of the sample mean is 36, what will be the standard deviation of the actual population?
Solution:
We know that variance of the sample mean is given by $\frac{\sigma^2}{n}$ where σ is the standard deviation of the actual population and n is the sample size.

By the given information, we have

$$\begin{aligned}
&\frac{\sigma^2}{n} = 36 \\
\Rightarrow &\frac{\sigma^2}{100} = 36 \\
\Rightarrow &\sigma^2 = 3600 \\
\Rightarrow &\sigma = 60
\end{aligned}$$

Therefore, standard deviation of the actual population is 60.

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4. A random sample of size 50 is collected from a population with a standard deviation of 5. Find the upper bound on the probability that the sample mean will be at least 10 away from the actual mean using the weak law of large numbers. Write your answer correct to three decimal places.
Solution:
Given: standard deviation of the population, $\sigma = 5$
Sample size, $n = 50$

To find: upper bound on $P(|\overline{X} - \mu| \geq 10)$ where $\overline{X}$ and μ are sample mean and population mean, respectively.

Now, by weak law of large number, we have

$$P(|\overline{X}-\mu| \geq \delta) \leq \frac{\sigma^2}{n\delta^2}$$
$$\Rightarrow P(|\overline{X}-\mu| \geq 10) \leq \frac{25}{100 \times 50}$$
$$\Rightarrow P(|\overline{X}-\mu| \geq 10) \leq 0.005$$

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5. A study says that the delivery time of pizzas has a standard deviation of 10 minutes. A pizza shop collected the data of some deliveries and their delivery time. The probability that the mean delivery time of this sample is at least $\sqrt{5}$ minutes away from the actual mean delivery time is at most $\frac{1}{5}$ as per the weak law of large numbers. What is the size of the sample?
Solution:
Let X denote the delivery time of pizzas.
Given that $\sigma = 10$
To find: size of the sample such that $P(|\overline{X}-\mu| \geq \sqrt{5}) \leq \frac{1}{5}$ ...(1).
By the weak law of large numbers, we have

$$P(|\overline{X}-\mu| \geq \delta) \leq \frac{\sigma^2}{n\delta^2}$$
$$\Rightarrow P(|\overline{X}-\mu| \geq \sqrt{5}) \leq \frac{100}{n \times 5} \qquad ...(1)$$

By equation (1) and (2), we have

$$\frac{1}{5} = \frac{100}{5n}$$
$$\Rightarrow n = 100$$

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6. A random sample of size 100 is collected from a normal population with mean μ and variance σ^2. Suppose the expected value and the variance of the sample mean is 50 and 0.25, respectively. Find the value of $\mu + 10\sigma$.
Solution:
Given, the population has mean μ and variance σ^2.
Let X denote the population. Then
$E[X] = \mu, \mathrm{Var}(X) = \sigma^2$
Sample size, $n = 100$.
Let $\overline{X} = \dfrac{X_1 + \ldots + X_{100}}{100}$ denote the sample mean.

We know that $E[\overline{X}] = E[X] = \mu$ and $\mathrm{Var}(\overline{X}) = \dfrac{\mathrm{Var}(X)}{n} = \dfrac{\sigma^2}{n}$

Since $E[\overline{X}] = 50, \implies \mu = 50$. Since $Var(\overline{X}) = 0.25 \implies \dfrac{\sigma^2}{n} = 0.25 \implies \sigma^2 = 25$

Now, the value of $\mu + 10\sigma = 50 + 10\sqrt{25} = 100$.

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7. Which among the following is a random variable? Select all that apply.

   (i) Population mean
   (ii) Sample mean
   (iii) Population variance
   (iv) Sample variance
   (v) Variance of the sample mean

   Solution:

   We know that sample mean, $\overline{X}$ is given by $\dfrac{X_1 + \ldots + X_n}{n}$. So, everytime we collect a sample, sample mean will give a different value and it will depend on the samples taken. So, it is a random variable.

   Similarly, the value of sample variance will depend on the samples taken, so it is a random variable.

   Expected value and variance of a population X is fixed, so they are not a random variable.

   We know that $\text{Var}(\overline{X}) = \dfrac{\sigma^2}{n}$, where σ^2 is population variance and n is the sample size, which are fixed, so the variance of sample mean is not a random variable.

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8. In a company, a testing of the product is done to check whether the product is defective or not. The probability that the product is defective is 0.2 independent of each other. 20 such products are randomly chosen for testing. Which of the following statements are true?
   Hint: Consider the product being non-defective as the success.

   (a) The expectation of a sample mean is 0.8.
   (b) The expectation of a sample mean is 16.
   (c) The variance of a sample mean is 0.008.
   (d) The variance of a sample mean is 0.16.

   Solution:

   Define a random variable X as

$$X = \begin{cases} 1, & \text{if the product is non-defective} \\ 0 & \text{otherwise} \end{cases}$$

Now, $P(X = 1) = 0.8$ and $P(X = 0) = 0.2$
Since X is Bernoulli(0.8), $E[X] = \mu = 0.8$ and $\text{Var}[X] = \sigma^2 = 0.8 \times 0.2 = 0.16$

A sample $X_1, \ldots, X_{20}$ is chosen for testing and $X_1, \ldots, X_{20} \sim iid\ X$.

Let $\overline{X}$ denotes the sample mean. We know that $E[\overline{X}] = \mu$ and $\text{Var}(\overline{X}) = \dfrac{\sigma^2}{n}$. Therefore,

$E[\overline{X}] = 0.8$.
Hence, option (a) is correct and (b) is incorrect.

And, $\text{Var}(\overline{X}) = \dfrac{\sigma^2}{n} = \dfrac{(0.8)(0.2)}{20} = 0.008$. Therefore, option (c) is correct and (d) is incorrect.

Use the following information to answer (9) and (10).

Suppose for the college president, two candidates are standing for the election, Candidate A and Candidate B. Of the students in the college, a proportion p will vote for Candidate A and a proportion $(1 - p)$ will vote for Candidate B. During the election, a random sample X_i's of students are independently interviewed for the candidate they are going to vote for, where i-th person interviewed will vote for Candidate A.

9. Find the minimum number of people who should be interviewed such that the probability that $\dfrac{\sum_{i=1}^{n} X_i}{n}$ is within 0.2 of the true p is at least 0.9, for $p = 0.4$?
Answer: 60

Solution:

Let the number of people who is interviewed be n. Each student has two choices: Candidate A and Candidate B.

Let X_i be defined as

$$X_i = \begin{cases} 1, & \text{if student } i \text{ votes for candidate } A \\ 0 & \text{otherwise} \end{cases}$$

Since X_i's follow Bernoulli(p) distribution, with $E[X_i] = p$ and $\text{Var}[X_i] = p(1 - p)$, we will have $E[\overline{X}] = p$ and $\text{Var}[\overline{X}] = \dfrac{p(1-p)}{n}$

, where $p = \dfrac{\text{Number of students voted for Candidate } A}{\text{Total number of students who voted}}$

Using Weak Law of Large Numbers, we know that

$$P(\mid \overline{X} - \mu \mid \leq c) \geq 1 - \frac{\sigma^2/n}{c^2}$$

Therefore,

$$P\left(\mid \frac{\sum_{i=1}^{n} X_i}{n} - p \mid \leq 0.2 \right) \geq 1 - \frac{(0.4)(0.6)/n}{0.2^2} \geq 0.9$$

Solving for $1 - \frac{(0.4)(0.6)/n}{0.2^2} \geq 0.9$, we get
$\frac{0.24}{0.04n} \leq 0.1 \implies n \geq 60$
Therefore, the minimum number of people who should be interviewed is 60.

10. Find the minimum number of people who should be interviewed such that the probability that $\frac{\sum_{i=1}^{n} X_i}{n}$ is within 0.2 of the true p is at least 0.95, for $p = 0.4$?
Answer: 120

Solution:

Using Weak Law of Large Numbers, we know that

$$P(\mid \overline{X} - \mu \mid \leq c) \geq 1 - \frac{\sigma^2/n}{c^2}$$

$$P\left(\mid \frac{\sum_{i=1}^{n} X_i}{n} - p \mid \leq 0.2 \right) \geq 1 - \frac{(0.4)(0.6)/n}{0.2^2} \geq 0.95$$

Solving for $1 - \frac{(0.4)(0.6)/n}{0.2^2} \geq 0.95$, we get
$\frac{0.24}{0.04n} \leq 0.05 \implies n \geq 120$
Therefore, the minimum number of people who should be interviewed is 120.